

Review

65%

13

2:01

Out of 20 points

Avg Time Per Question

Your Answers:

Vignette

You are a junior analyst at an asset management firm. Your supervisor asks you to analyze the return drivers for one of the firm’s portfolios. She asks you to construct a regression model of the portfolio’s monthly excess returns (RET) against three factors: the market excess return (MRKT), a value factor (HML), and the monthly percentage change in a volatility index (VIX). You collect the data and run the regression. After completing the first regression (Model 1), you review the ANOVA results with your supervisor.

Then, she asks you to create two more models by adding two more explanatory variables: a size factor (SMB) and a momentum factor (MOM). Your three models are as follows:

- Model 1: $RET_i = b_0 + b_{MRKT}MRKT_i + b_{HML}HML_i + b_{VIX}VIX_i + \epsilon_i$.
- Model 2: $RET_i = b_0 + b_{MRKT}MRKT_i + b_{HML}HML_i + b_{VIX}VIX_i + b_{SMB}SMB_i + \epsilon_i$.
- Model 3: $RET_i = b_0 + b_{MRKT}MRKT_i + b_{HML}HML_i + b_{VIX}VIX_i + b_{SMB}SMB_i + b_{MOM}MOM_i + \epsilon_i$.

The regression statistics and ANOVA results for the three models are shown in Exhibit 1, Exhibit 2, and Exhibit 3.

Exhibit 1
ANOVA Table for Model 1

$RET_i = b_0 + b_{MRKT}MRKT_i + b_{HML}HML_i + b_{VIX}VIX_i + \epsilon_i$

Regression Statistics			Coefficient	Std. Error	t-Stat.	P-Value
Multiple R	0.907	Intercept	−0.999	0.414	−2.411	0.018
R-Squared	0.823	MRKT	1.817	0.124	14.683	0.000
Adjusted R-Sq.	0.817	HML	0.489	0.118	4.133	0.000
Standard Error	3.438	VIX	0.037	0.018	2.122	0.037
Observations	96.000					

ANOVA

	<i>Df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	3	5058.430	1686.143	142.628	0.000
Residual	92	1087.618	11.822		
Total	95	6146.048			

Exhibit 2

ANOVA Table for Model 2

$$RET_i = b_0 + b_{MRKT}MRKT_i + b_{HML}HML_i + b_{VIX}VIX_i + b_{SMB}SMB_i + \epsilon_i$$

<i>Regression Statistics</i>			<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Stat.</i>	<i>P-Value</i>
Multiple R	0.923	Intercept	−0.820	0.383	−2.139	0.035
R-Squared	0.852	MRKT	1.649	0.121	13.683	0.000
Adjusted R-Sq.	0.846	HML	0.434	0.109	3.970	0.000
Standard Error	3.161	VIX	0.025	0.016	1.516	0.133
Observations	96.000	SMB	0.563	0.133	4.223	0.000

ANOVA

	<i>Df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	4	5236.635	1309.159	131.000	0.000
Residual	91	909.413	9.994		
Total	95	6146.048			

Exhibit 3

ANOVA Table for Model 3

$$RET_i = b_0 + b_{MRKT}MRKT_i + b_{HML}HML_i + b_{VIX}VIX_i + b_{SMB}SMB_i + b_{MOM}MOM_i + \epsilon_i$$

<i>Regression Statistics</i>			<i>Coefficient</i>	<i>Std. Error</i>	<i>t-Stat.</i>	<i>P-Value</i>
Multiple R	0.923	Intercept	−0.823	0.385	−2.136	0.035

$$RET_i = b_0 + b_{MRKT}MRKT_i + b_{HML}HML_i + b_{VIX}VIX_i + b_{SMB}SMB_i + b_{MOM}MOM_i + \varepsilon_i$$

Regression Statistics			Coefficient	Std. Error	t-Stat.	P-Value
R-Squared	0.852	MRKT	1.719	0.280	6.130	0.000
Adjusted R-Sq.	0.844	HML	0.412	0.138	2.989	0.004
Standard Error	3.177	VIX	0.026	0.017	1.532	0.129
Observations	96.000	SMB	0.553	0.139	3.987	0.000
		MOM	-0.067	0.242	-0.276	0.783

ANOVA

	Df	SS	MS	F	Significance F
Regression	5	5237.402	1047.480	103.751	0.000
Residual	90	908.647	10.096		
Total	95	6146.048			

Your supervisor asks for your assessment of the model that provides the best fit as well as the model that is best for predicting values of the monthly portfolio return. So, you calculate Akaike's information criterion (AIC) and Schwarz's Bayesian information criterion (BIC) for all three models, as shown in Exhibit 4.

Exhibit 4

Goodness-of-Fit Measures

	AIC	BIC
Model 1	241.03	251.29
Model 2	225.85	238.67
Model 3	227.77	243.16

1 Multiple Choice 1 / 1 point

Determine which one of the following reasons for the change in adjusted R^2 from Model 2 to Model 3 is *most likely* to be correct.

- ☒ A. Adjusted R^2 decreases since adding MOM does not improve the overall explanatory power of Model 3.
- ☐ B. Adjusted R^2 increases since adding SMB improves the overall explanatory power of Model 2.

- ☐ C. Adjusted R^2 decreases since adding MOM improves the overall explanatory power of Model 3.

Feedback

A is correct. Adjusted R^2 in Model 3 decreases to 0.844 from 0.846 in Model 2. Model 3 includes all independent variables from Model 2, while adding MOM. Adding variables to a regression model always either increases R^2 or causes it to stay the same. But adjusted R^2 only increases if the new variable meets a threshold of significance, $|t\text{-statistic}| > 1$. MOM does not meet this threshold, indicating it does not improve the overall explanatory power of Model 3.

2 Multiple Choice 0 / 1 point

Identify the model that provides the best fit.

☒ A. Model 1

Correct Answer: **B. Model 2**

☐ B. Model 2

☐ C. Model 3

Feedback

B is correct. BIC is the preferred measure for determining which model provides the best fit, and a lower BIC is better. Since Model 2 has the lowest BIC value, it provides the best fit among the three models.

3 Multiple Choice 0 / 1 point

Identify the model that should be used for prediction purposes.

☒ A. Model 1

Correct Answer: **B. Model 2**

☐ B. Model 2

☐ C. Model 3

Feedback

B is correct. AIC is the preferred measure for determining the model that is best used for prediction purposes. As with BIC, a lower AIC is better. Model 2 also has the lowest AIC value among the three models; thus, it should be used for prediction purposes.

4 Multiple Choice 0 / 1 point

Calculate the predicted RET for Model 3 given the assumed factor values: MRKT = 3, HML = -2, VIX = -5, SMB = 1, MOM = 3.

☐ A. 3.732



☒ B. 3.992

Correct Answer:

A. 3.732

☐ C. 4.555

Feedback

A is correct. The regression equation for Model 3 is

$$\text{RET} = -0.823 + 1.719\text{MRKT} + 0.412\text{HML} + 0.026\text{VIX} + 0.553\text{SMB} - 0.067\text{MOM}.$$

Using the assumed values for the independent variables, we have

$$\text{RET} = -0.823 + (1.719)(3) + (0.412)(-2) + (0.026)(-5) + (0.553)(1) - (0.067)(3) = 3.732.$$

5 Multiple Choice 1 / 1 point

Calculate the joint F -statistic and determine whether SMB and MOM together contribute to explaining RET in Model 3 at a 1% significance level (use a critical value of 4.849).

☐ A. 2.216, so SMB and MOM together do not contribute to explaining RET



☒ B. 8.863, so SMB and MOM together do contribute to explaining RET

☐ C. 9.454, so SMB and MOM together do contribute to explaining RET

Feedback

B is correct. To determine whether SMB and MOM together contribute to the explanation of RET, at least one of the coefficients must be non-zero. So, $H_0: b_{\text{SMB}} = b_{\text{MOM}} = 0$ and $H_a: b_{\text{SMB}} \neq 0$ and/or $b_{\text{MOM}} \neq 0$.

We use the F -statistic, where

$$F = \frac{\text{SSR}_{\text{restricted}} - \text{SSR}_{\text{unrestricted}} / q}{\text{SSR}_{\text{unrestricted}} / (n - k - 1)},$$

with $q = 2$ and $n - k - 1 = 90$ degrees of freedom. The test is one-tailed, right side, with $\alpha = 1\%$, so the critical F -value is 4.849.

Model 1 does not include SMB and MOM, so it is the restricted model. Model 3 includes all of the variables of Model 1 as well as SMB and MOM, so it is the unrestricted model.

Using data in Exhibit 1 and Exhibit 3, the joint F -statistic is calculated as

Since $8.863 > 4.849$, we reject H_0 . Thus, SMB and MOM together do contribute to the explanation of RET in Model 3 at a 1% significance level.

Vignette

Anna Gabor is an analyst with an investment management firm. She is investigating the impact of leverage on annual ROE in the firm's Eurozone equities fund ("the fund").

She collects data on the 72 stocks in the fund, including company size as an independent variable.

Exhibit 1 shows the results from estimating the model $R_i = b_0 + b_1(\text{Size}_i) + b_2(D/E_i) + \varepsilon_i$

Where:

- R_i is the annual ROE for company i in the last calendar year in numbers (0.05 is 5%);
- Size_i is the natural log of the market capitalization of company i (millions of Euro) on 31 December;
- D/E_i is the debt/equity ratio of company i on 31 December;
- b_0 is the intercept of the equation;
- b_1, b_2 are the slope coefficients for the independent variables Size_i and D/E_i respectively;
- ε_i is the error term.

Exhibit 1

	Coefficient	Standard Error
Intercept	0.2248	---
Size_i	-0.012	---
D/E_i	-0.0172	0.00986
ANOVA	Degrees of freedom	Sum of squares
Regression	2	0.0960
Residual	69	0.5704

Exhibit 2 shows selected values of Student's t distribution for 69 degrees of freedom.

Exhibit 2
Selected Values of Student's t distribution

Area in Right Tail	t Value (69 degrees of freedom)
0.05	1.6670
0.025	1.9950

Gabor expects to find that both the debt/equity ratio and size are negatively related to ROE. She considers a relationship meaningful when it is statistically significant at the 0.05 level. As part of her analysis Gabor compares the predictions of her model with the actual ROE for two stocks in the portfolio which have the same debt ratio: A and B, which have market capitalizations EUR 100 million and EUR 200 million respectively.

Gabor shares her results with a colleague, Eva Szabo, who warns that the variance of the error terms might not be constant, but positively correlated with D/E and Size. Szabo states that such a flaw would lead to the F- and t-statistics being biased and failing to reject the null hypothesis when it is false.

As an additional goal Gabor investigates the impact of qualitative governance factors. She is considering adding two dummy variables, to the model: D_1 and D_2 . D_1 is 1 for firms with two or more female board members, and otherwise 0. The value of D_2 is 1 if the audit committee of the board includes only independent directors, and otherwise 0. Gabor's hypothesis is that gender diversity is positively related to ROE and that an independent audit committee slows the rate at which increasing leverage affects ROE.

Finally, Gabor wishes to test the hypothesis that whether or not a stock in the fund is listed on a US (as well as European) exchange is determined by the same independent variables as above (including the governance factors). She employs a logistic regression with the dependent variable of A, which takes on a value of 1 for a company listed on a US exchange and otherwise 0. For Tekdat, a company in the fund which does not have gender diversity on its board, the probability, based on the model, of being listed on a US exchange is 0.65. The regression coefficient of D_1 in the logistic model is 0.80 and is statistically significant.

6 Multiple Choice 1 / 1 point

Based on Exhibits 1, 2 and Gabor's expectations, which is the *best* null hypothesis and conclusion regarding D/E:

☒ $b_2 \geq 0$; reject the null hypothesis.

Correct Answer Feedback: Correct because the null hypothesis is correctly formulated and we can reject it. Gabor expects to find that D/E is negatively related to the firm's ROE. If she is right, the regression coefficient for D/E, b_2 , will be less than zero and statistically significant.

The null hypothesis supposes that the 'suspected' condition is not true, so the null hypothesis should state the variable is greater than or equal to zero. Therefore, we should be conducting a one-tailed test.

The t-Statistic for b_2 is given by: $-0.0172/0.0098 = -1.7551$

From Exhibit 2 we see that the critical value for a one-tailed test with 69 degrees of freedom at the 0.05 significance level is 1.667.

So, we should reject the null hypothesis if the t-statistic is less than -1.667

Since $-1.7551 < -1.667$ we should indeed reject the null hypothesis.

☐ $b_2 = 0$; cannot reject the null hypothesis.

Incorrect Answer Feedback: Incorrect because the null hypothesis is incorrectly formulated.

Gabor expects to find a negative relationship between D/E and ROE that is also statistically significant. This suggests an expectation that b_2 is less than zero, leading to a null hypothesis that b_2 is greater than or equal to zero. Hence, we need to conduct a one-tailed test.

Instead, the candidate has specified a null hypothesis for a two-tailed test. This would be correct if Gabor's expectation was just that there was a statistically significant relationship between D/E and ROE, remaining neutral as to whether the relationship was positive or negative.

With this null hypothesis, the critical value for a t-test would be 1.995, as per Exhibit 2.

The t-Statistic for b_2 is given by: $-0.0172/0.0098 = -1.7551$

Since the absolute value of this t-statistic is less than the critical value we would not be able to reject the null hypothesis. This conclusion would be correct for a two-tailed test.

So, by incorrectly specifying the null hypothesis, the candidate has also arrived at the wrong conclusion.

☐ $b_2 \geq 0$; cannot reject the null hypothesis.

Incorrect Answer Feedback: Incorrect because, though the null hypothesis is correctly formulated, the conclusion is incorrect.

The t-Statistic for b_2 is given by: $-0.0172/0.0098 = -1.7551$

From Exhibit 2 we see that the critical value for a one-tailed test with 69 degrees of freedom at the 0.05 significance level is 1.667.

Since $-1.7551 < -1.667$ we can reject the null hypothesis. Hence the conclusion is incorrect.

A candidate who used the critical value for a one-tailed test in error would compare the t-statistic of -1.7551 with -1.995 and would have concluded that the null hypothesis cannot be rejected.

7 Multiple Choice 0 / 1 point

Based on Gabor's regression, the amount that ROE for Company A exceeds that of Company B (in percentage points) is *closest* to:

☐ 0.83.

Correct Answer Feedback: Correct, because this reflects the right application of the results of the regression.

The coefficient of the independent variable Size_i is $b_1 = -0.012$. This implies a negative relationship between the log of market capitalization and ROE.

The prediction of ROE, according to the regression, is given by the equation:

$$R_i = b_0 + b_1(\text{Size}_i) + b_2(D/E_i)$$

Or, reading the coefficients from Exhibit 1:

$$R_i = 0.2248 - 0.012(\text{Size}_i) - 0.0172(D/E_i)$$

We are told that the debt/equity ratio D/E is the same for both companies, therefore $0.2248 - 0.0172(D/E_i)$ is the same amount for both companies A and B; let us say that

$$0.2248 - 0.0172(D/E_i) = K \text{ (a constant) for both A and B}$$

$$\text{Then } R_i = K - 0.012(\text{Size}_i)$$

The market cap of Company A is EUR 100 million, therefore $\text{Size}_A = \ln(100)$

The market cap of Company B is EUR 200 million, therefore $\text{Size}_B = \ln(200) = \ln(2) + \ln(100)$

ROE for Company A is $K - 0.012 \times \ln(100)$ and the ROE for Company B is: $K - 0.012 \times \ln(100) - 0.012 \times \ln(2)$

So, ROE for Company A exceeds that for Company B by:

$$0.012 \times \ln(2) = 0.008318 \text{ or approximately } 0.83 \text{ percentage points.}$$

☐ 1.20.

✗

Incorrect Answer Feedback: Incorrect because this answer reflects an erroneous application of the regression coefficient.

The candidate has multiplied the arithmetic difference between the market cap of Companies A and B by the coefficient of $Size_i$, giving an absolute amount of:

$$\text{EUR } 100 \text{ million} \times 0.012 = 1.2$$

The candidate has failed to take into account that $Size_i$ is the coefficient of the natural logarithm of the market capitalization. They have also omitted to scale the result of the calculation to percentage points.

A candidate who applied the wrong choice of regression coefficient i.e. -0.0172 would arrive at the same incorrect answer. This is the coefficient of D/E_i , and not $Size_i$.

This again implies a negative relationship between the log of market capitalization and ROE.

Proceeding to apply the regression equation, the candidate would arrive at the following answer for the absolute value of the difference:

$$0.0172 \times \ln(2) = 0.01192 \text{ or approximately } 1.19 \text{ percentage points which is closest to } 1.2$$

☐ 6.36.

Incorrect Answer Feedback: Incorrect because this answer reflects an erroneous application of the regression coefficient.

The candidate has multiplied the coefficient of $Size_i$ by the natural logarithm of the total market cap of Company B, i.e. EUR 200. The calculation of the absolute value is then given by:

$$\ln(200) \times 0.012 = 0.06357 \text{ or approximately } 6.36\%$$

Vignette

Andrew Omandi works as a senior analyst at investment firm CPSR Partners. He and junior analysts, Emmanuel Katangole, Takasongo Kasongo and Peter Mensah, have been tasked with investigating the use of multi factor models to help explain portfolio returns. After conducting some research, they identify a three-factor model described below, and are meeting to finalize their presentation to the firm's investment committee.

$$R_{it} - R_{ft} = \text{intercept}_{it} + B_M(R_{Mt} - R_{ft}) + B_{SMB}(SMB_t) + B_{HML}(HML_t) + e_{it}$$

Where:

R_{it} = portfolio return

R_{ft} = risk free rate, one-month T-bill return

B_M = market regression coefficient

R_{Mt} = return on market portfolio

$R_{Mt} - R_{ft}$ = market risk premium

B_{SMB} = SMB regression coefficient

SMB_t = return difference between small cap stocks and large cap stocks (size premium)

B_{HML} = HML regression coefficient

HML_t = return difference between high book to market stocks and low book to market stocks (value premium)

e_{it} = error term

Omandi states: "This model indicates that the main factors driving expected portfolio excess returns are premiums for market risk ($R_{Mt} - R_{ft}$), size (SMB_t) and value (HML_t). I also believe that there is a positive relationship between portfolio excess return and each of the independent variables, market risk, size and value premiums. Based on this we can formulate the following hypotheses:

Hypothesis 1

$H_o: B_M = 0$

$H_a: B_M \neq 0$

Hypothesis 2

$$H_o: B_{SMB} \leq 0$$

$$H_a: B_{SMB} > 0$$

Hypothesis 3

$$H_o: B_{HML} > 0$$

$$H_a: B_{HML} \leq 0$$

The analysts test the model and the regression results of excess portfolio returns on Mkt-Rf, SMB and HML are presented below in Exhibit 1.

Exhibit 1

Regression Statistics	
Multiple R ²	0.6235
Standard error	0.0774
Observations	60

ANOVA	Degrees of Freedom (df)	Sum of Squares (SS)	Mean Squares (MSS)	F	Significance F
Regression	3	0.2138	0.0713	11.871	0
Residual	56	0.3362	0.0060	---	---
Total	59	0.55	---	---	---

	Coefficient	Standard Error	t-Statistic	p-value
Intercept	0.09219	0.0109	8.4577	<0.00001
Mkt-Rf	0.01348	0.00248	5.4354	<0.00001
SMB	0.0077	0.004534	1.6982	0.04829
HML	0.0043	0.00323	1.3312	0.09445

Kasango states that it is important to emphasize that the multiple linear regression model makes a number of assumptions, three of which are:

- Assumption 1: The regression residuals are normally distributed
- Assumption 2: The variance of the regression residuals is the same for all observations.
- Assumption 3: The regression residuals are correlated across observations.

Katangole asks how one can assess the goodness of fit of the estimated regression to the data.

Mensah responds, "One measure, the R² can be defined as the ratio of the variation in the dependent variable explained by the independent variables to the total variation of the dependent variable. However, R² stays the same or increases when independent variables are added to the regression. A better measure is the adjusted R² which does not automatically increase when independent variables are added to the regression."

8 Multiple Choice 0 / 1 point

Which of Omandi's three hypotheses has been correctly specified?

☒ Hypothesis 1



Incorrect Answer Feedback: Incorrect. Omandi indicates he believes there is positive relationship between portfolio excess return and each of the independent variables, market risk, size and value premiums. Hypothesis 1 does not test for this expectation. Here, if the null hypothesis is rejected, and the alternate hypothesis is accepted, we can only infer that there is a relationship and it may be positive or negative, not that the relationship is positive. The correct specification of the null hypothesis is

$$H_0: B_M \leq 0$$

$$H_a: B_M > 0$$

We want to see if we can reject the null that B_M less than or equal to zero and accept the alternate that it is greater than zero, not reject the null that B_M is equal to zero and accept the alternate that it is not equal to zero.

☐ Hypothesis 2

Correct Answer Feedback: Correct. Omandi states that this model indicates that the main factors driving expected portfolio excess returns are premiums for market risk ($R_{Mt} - R_{ft}$), size (SMB_t) and value (HML_t). He expects that higher premiums for market risk, size and value will drive higher expected excess portfolio returns. That is, in each case we will want to reject the null hypothesis that the slope coefficients are less than or equal to zero and accept the alternate hypothesis that they are greater than zero. Based on this the following three hypotheses can be tested

Hypothesis 1

$$H_0: B_M \leq 0$$

$$H_a: B_M > 0$$

Hypothesis 2

$$H_0: B_{SMB} \leq 0$$

$$H_a: B_{SMB} > 0$$

Hypothesis 3

$$H_0: B_{HML} \leq 0$$

$$H_a: B_{HML} > 0$$

☐ Hypothesis 3

Incorrect Answer Feedback: Incorrect. Omandi indicates he believes there is a positive relationship between portfolio excess return and each of the independent variables, market risk, size and value premiums. Hypothesis 3 does not test for this expectation. Here if the null hypothesis is rejected, and the alternate hypothesis is accepted, we can only infer that there is no relationship or that the relationship is negative, not that the relationship is positive. The correct specification is:

$$H_0: B_{HML} \leq 0$$

$$H_a: B_{HML} > 0$$

This is because we want to see if we can reject the null hypothesis that B_{HML} is negative or less than zero, not reject the null that B_{HML} is greater than zero.

9 Multiple Choice 0 / 1 point

Is Mensah's response to Katangole about the goodness of fit of the estimated regression to the data *most likely* correct?

☐ Yes

Correct Answer Feedback: Correct. In his response to Katangole, Mensah is correct about multiple R^2 and adjusted R^2 . The R^2 provides an indication of how much of the variation in the dependent variable, excess portfolio returns, is explained by the independent variables, $R_{Mt}-R_{ft}$, SMB_t and HML_t . The multiple R^2 here is 0.6235, so there is a possibility that they may have omitted independent variables from the equation. Also, However, in a multiple linear regression R^2 can be increased simply by adding independent variables - it may stay the same but will never decrease with added variables., and this happens when the new independent variable is not a linear combination of the other independent variables and is at least slightly correlated with the dependent variable. The adjusted R^2 is a better measure of goodness of fit in a multiple regressions because it does not automatically increase when independent variables are added to the regression.

☐ No, he is incorrect about R^2

Incorrect Answer Feedback: Incorrect as Mensah is correct about multiple R^2 . The R^2 provides an indication of how much of the variation in the dependent variable, excess portfolio returns, is explained by the independent variables, $R_{Mt}-R_{ft}$, SMB_t and HML_t . The multiple R^2 here is 0.6235, so there is a possibility that they may have omitted independent variables from the equation. Also, However, in a multiple linear regression R^2 can be increased simply by adding independent variables, and this happens when the new independent variable is not a linear combination of the other independent variables and is at least slightly correlated with the dependent variable.

☒ No, he is incorrect about adjusted R^2

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Incorrect Answer Feedback: Incorrect as Mensah is correct about adjusted R^2 . The adjusted R^2 is a better measure of goodness of fit in a multiple regressions because it does not automatically increase when independent variables are added to the regression.

10 Multiple Choice 1 / 1 point

Based on Exhibit 1 and given expected values for the independent variables of: $(R_{Mt}-R_{ft}) = 0.0545$, $SMB_t = .0423$, and $HML_t = .0326$, the expected excess portfolio return is *closest* to:

☒ 0.0934

Correct Answer Feedback: Correct.

$$R_{it} - R_{ft} = \text{intercept}_{it} + B_M(R_{Mt}-R_{ft}) + B_{SMB}(SMB_t) + B_{HML}(HML_t) + e_{it}$$

$$R_{it} - R_{ft} = .09219 + .01348 \times .0545 + .0077 \times .0423 + .0043 \times .0326 \\ = .093391$$

☐ 0.1177

Incorrect Answer Feedback: Incorrect. As this response was derived by simply adding the parameter values

$$.09219 + .01348 + .0077 + .0043 = .11767$$

☐ 0.1294

Incorrect Answer Feedback: Incorrect. As this response was derived by adding the expected values for the independent variables.

$$.0545 + .0423 + .0326 = .1294$$

Vignette

Matthew Markham is a recently hired analyst at an equity research firm. Markham joins several other new employees for a training session hosted by Alexandra Garcia, a senior analyst, on the firm's approach to utilizing multiple regression in their portfolio and investment analysis.

To open the training session, Garcia explains the relationship between the dependent and independent variables in a regression model. She explains the multiple linear regression model and asks the analysts to demonstrate their understanding by confirming the assumptions for the classical model. The analysts provide the following responses about the assumptions:

- Assumption 1: The regression residuals are normally distributed.
- Assumption 2: The independent variables are not random.
- Assumption 3: The regression residuals are correlated across observations.

Garcia shares an example of a regression model completed at the company, which was based on 650 observations and 11 independent variables, and advises that a specific variable with a coefficient of 1.25 has a t -statistic of 2.39. She shares that the critical values for a two-sided t -test is 1.96 at the 0.05 significance level. Based on this information, she asks Markham to confirm if they should reject or fail to reject the null hypothesis.

In completing their review of the previous model, Garcia cautions the analysts to be aware of the various types of uncertainty that can arise when predicting the dependent variable using a linear regression model. She identifies two errors that result in the forecast of the dependent variable having a standard error larger than the standard error of the regression:

- Error 1: Model error
- Error 2: Sampling error

Garcia advises that the firm prefers to use the F -test to evaluate the overall significance of a multiple regression model and asks Markham to calculate the F -statistic for a recent model that is summarized in Exhibit 1.

Exhibit 1

ANOVA	Degrees of freedom	Sum of Squares	Mean squares
Regression	2	1,924.01	962.01
Residual	122	1,683.87	13.80
Total	124	3,607.88	--
Residual standard error	--	--	0.798
Multiple R^2	--	--	0.815
Observations	--	--	125

11 Multiple Choice 1 / 1 point

In Garcia's regression model example, the *most* accurate response from Markham is to:

- ☒ reject the null hypothesis based on the t -statistic.
Correct Answer Feedback: Correct because the t -statistic of 2.39 exceeds the critical value for this t -test of 1.96 at the 0.05 significant level, which means that the null hypothesis should be rejected.
- ☐ fail to reject the null hypothesis based on the t -statistic.
Incorrect Answer Feedback: Incorrect because the t -statistic of 2.39 exceeds the critical value for this t -test of 1.96 at the 0.05 significant level, which means that the null hypothesis should be rejected.
- ☐ fail to reject the null hypothesis based on the value of the co-efficient.
Incorrect Answer Feedback: Incorrect because the value of the coefficient is not the basis for rejecting or failing to reject the null hypothesis.

12 Multiple Choice 1 / 1 point

Is Garcia correct about the errors affecting the standard error of the forecast of the dependent variable?

- ☒ Yes



Correct Answer Feedback: Correct because when predicting the dependent variable using a linear regression model, we encounter two types of uncertainty: uncertainty in the regression model itself, as reflected in the standard error of estimate, and uncertainty about the estimates of the regression model's parameters. In any regression estimation, there are residuals because not all observations lie on the estimated line. This is basic uncertainty in the model—the model error—which is the stochastic part of the model that involves the regression residual, ϵ_i . When the independent variables themselves are forecasts and thus out-of-sample predictions, there is an added source of error arising from errors associated with forecasting the independent variables. In such cases, the forecast error for the dependent variable is dependent on how well the independent variables (X_{1f} , X_{2f} , ..., X_{kf}) were forecasted, hence introducing sampling error. The combined effect of the model error and the sampling error results in a standard error of the forecast for $\hat{Y}_f$ that is larger than the standard error of the regression. This larger forecast error results in a prediction interval for the dependent variable that is wider than the within-sample error.

- ☐ No, she is incorrect about Error 1

Incorrect Answer Feedback: Incorrect because Error 1, model error, works to increase the forecast interval (standard error) for the dependent variable. When predicting the dependent variable using a linear regression model, we encounter two types of uncertainty: uncertainty in the regression model itself, as reflected in the standard error of estimate, and uncertainty about the estimates of the regression model's parameters.

It is the combined effect of the model error and the sampling error that results in a standard error of the forecast for $\hat{Y}_f$ that is larger than the standard error of the regression. This larger forecast error results in a prediction interval for the dependent variable that is wider than the within-sample error.

- ☐ No, she is incorrect about Error 2

Incorrect Answer Feedback: Incorrect because Error 2, sampling error, works to increase the forecast interval for the dependent variable. When predicting the dependent variable using a linear regression model, we encounter two types of uncertainty: uncertainty in the regression model itself, as reflected in the standard error of estimate, and uncertainty about the estimates of the regression model's parameters.

It is the combined effect of the model error and the sampling error that results in a standard error of the forecast for $\hat{Y}_f$ that is larger than the standard error of the regression. This larger forecast error results in a prediction interval for the dependent variable that is wider than the within-sample error.

13 Multiple Choice 1 / 1 point

Based on Exhibit 1, the F -statistic that Markham is asked to calculate is *closest* to:

- ☐ 34.9.

Incorrect Answer Feedback: Incorrect because it uses MSS (mean sum of squares) instead of RSS (regression sum of squares).

This result is incorrectly calculated by:

where

MSS = mean sum of squares

k = degrees of freedom

SSE = sum of square errors

n = number of observations

$F = (962.01 / 2) / (1,683.87 / [125 - (2 + 1)])$

$F = 34.9.$

- ☐ 53.4.

Incorrect Answer Feedback: Incorrect because this response incorrectly switches RSS (1,924.01) and SSE (1,683.87) as a calculation error.

This result is calculated by:

where

RSS = regression sum of squares

k = degrees of freedom

SSE = sum of square errors

n = number of observations

$$F = (1,683.87 / 2) / (1,924.01 / [125 - (2 + 1)])$$

$$F = 53.4.$$



69.7.

Correct Answer Feedback: Correct because the F -statistic would be calculated as:

where

RSS = regression sum of squares

k = degrees of freedom

SSE = sum of square errors

n = number of observations

$$F = (1,924.01 / 2) / (1,683.87 / [125 - (2 + 1)])$$

$$F = 69.7.$$

Vignette

Alwyn Rivera is a portfolio manager at Elite Investments (EI) based in the US. Rivera has a shortlist of 50 US public firms and wants to better understand the factors that may explain the cross-sectional variation in these stocks' returns. Based on the finance literature, Rivera believes that firm size and book-to-market ratio are important factors that need to be included in his model. He also thinks that being included in the Dow Jones Industrial Average Index (DJIA) can help a firm get more media attention and potentially attract more investors, which in turn may affect a stock's return.

To test his theory, Rivera collects the annual returns (r) of these 50 stocks in the year that just ended, as well as their market capitalizations (Size, in billions of dollars) and book-to-market ratios (BMRatio) at the end of the previous year. He also creates a dummy variable (Dow) that is coded 1 if a stock is included in the DJIA and 0 otherwise, and then runs the following regression:

$$r_i = b_0 + b_1 \ln \text{Size}_i + b_2 \text{BMRatio}_i + b_3 \text{Dow}_i + \varepsilon_i$$

where $\ln \text{Size}_i$ is the natural log of the market capitalization of company i in billions of dollars and Dow_i is the dummy variable. The output from the regression model is shown in Exhibit 1.

Exhibit 1

	Coefficient	t -statistic	p -value
Intercept	-0.6483	-3.8773	0.0003
Size	0.1777	5.1125	<0.0001
BMRatio	0.0433	2.9346	0.0052

InDow	-0.1641	-1.8362	0.0728
R^2	0.3700		
Adjusted R^2	0.3289		
Standard Error	0.2827		
Observations	50		

ANOVA	Degrees of freedom	Sum of squares	Mean squares	F-statistic	Significance F
Regression	3	2.1593	0.7198	9.006	<0.0001
Residual	46	3.6766	0.0799		
Total	49	5.8359			

Rivera shows his model and results to a summer intern, and asks whether he should use the reported R^2 or adjusted R^2 to measure the goodness of fit of his model. The intern replies, "Adjusted R^2 would be more appropriate in this case because the model has more than one independent variable. Adjusted R^2 is adjusted for degrees of freedom and nondecreasing in the number of independent variables."

14 Multiple Choice 1 / 1 point

Based on the results in Exhibit 1, which of the following interpretations regarding the coefficient of the variable BMRatio is *most* accurate?

- ☐ A stock's annual return is expected to increase by 1% for each 0.043% increase in its book-to-market ratio.

Incorrect Answer Feedback: Incorrect because the slope coefficients in a multiple regression are known as partial regression coefficients or partial slope coefficients and need to be interpreted with care. For example, suppose the coefficient on X_1 in a regression with the second independent variable was 0.60. We would interpret 0.60 as the expected increase in Y for a 1-unit increase X_1 holding the second independent variable constant. So, the coefficient of 0.043 for the BMRatio variable would be interpreted as a stock's annual return is expected to increase by 0.043% for each 1% increase in its book-to-market ratio holding MCap and InDow variables constant.

- ☒ Holding MCap and InDow variables constant, a stock's annual return is expected to increase by 0.043% for each 1% increase in its book-to-market ratio.

Correct Answer Feedback: Correct because the slope coefficients in a multiple regression are known as partial regression coefficients or partial slope coefficients and need to be interpreted with care. For example, suppose the coefficient on X_1 in a regression with the second independent variable was 0.60. We would interpret 0.60 as the expected increase in Y for a 1-unit increase X_1 holding the other second independent variables constant. So, the coefficient of 0.043 for the BMRatio variable would be interpreted as a stock's annual return is expected to increase by 0.043% for each 1% increase in its book-to-market ratio holding MCap and InDow variables constant. From the reading: Regarding interpretation of slope coefficients, as noted earlier, in multiple regression, the slope coefficient for a given independent variable is the expected change in the dependent variable for a one-unit change in that independent variable with all other independent variables remaining constant.

- ☐ Assuming MCap and InDow variables are both zero, a stock's annual return is expected to increase by 0.043% for each 1% increase in its book-to-market ratio.

Incorrect Answer Feedback: Incorrect because the slope coefficients in a multiple regression are known as partial regression coefficients or partial slope coefficients and need to be interpreted with care. For example, suppose the coefficient on X_1 in a regression with the second independent variable was 0.60. We would interpret 0.60 as the expected increase in Y for a 1-unit increase X_1 holding the second independent variable constant. So, the coefficient of 0.043 for the BMRatio variable would be interpreted as a stock's annual return is expected to increase by 0.043% for each 1% increase in its book-to-market ratio holding MCap and InDow variables constant. When MCap and InDow variables are both zero, the model will become a simple regression and the coefficient on the BMRatio variable would be different from 0.043.

15 Multiple Choice 0 / 1 point

Using two-tailed t -tests to determine if the coefficients of the independent variables are equal to zero at the 0.05 significance level, the null hypotheses are *most likely* rejected for the coefficients of:

- ☐ Size and InDow only.

Incorrect Answer Feedback: Incorrect because the p -value for the coefficient of the independent variable InDow is 0.0728, which is greater than 0.05, we cannot reject the null hypothesis that the coefficient of the InDow variable is zero at the 0.05 significance level. For each regression coefficient, the p -value would be the smallest level of significance at which we can reject a null hypothesis that the population value of the coefficient is 0, in a two-sided test. The p -values for the coefficients of Size and BMRatio are both less than 0.05. We can reject the null that the coefficients of Size and BMRatio; variables are equal to zero at the 0.05 significance level.

- ☒ Size and BMRatio only.

Correct Answer Feedback: Correct because the p -values for the coefficients of Size and BMRatio are both less than 0.05, and the p -value for the coefficient of the independent variable InDow is 0.0728, which is greater than 0.05. We can only reject the null hypotheses that the coefficients of the Size and BMRatio variables are equal to zero at the 0.05 significance level, but not the InDow variable. For each regression coefficient, the p -value would be the smallest level of significance at which we can reject a null hypothesis that the population value of the coefficient is 0, in a two-sided test.

- ☐ InDow and BMRatio only.

Incorrect Answer Feedback: Incorrect because the p -value for the coefficient of the independent variable InDow is 0.0728, which is greater than 0.05. The p -values for the coefficients of Size and BMRatio are both less than 0.05. We can reject the null that the coefficients of Size and BMRatio variables are equal to zero at the 0.05 significance level, but not the InDow variable. For each regression coefficient, the p -value would be the smallest level of significance at which we can reject a null hypothesis that the population value of the coefficient is 0, in a two-sided test.

16 Multiple Choice 1 / 1 point

The predicted annual return for a stock that is included in the DJIA, has a market capitalization of \$52 billion and a book-to-market ratio of 0.45 is *closest* to:

- ☒ -9.08%.

Correct Answer Feedback: Correct because the estimated model is $R_i = -0.6483 + 0.1777 \times \text{Size}_i + 0.0433 \times \text{BMRatio}_i - 0.1641 \times \text{Dow}_i$. When $\text{Size}_i = \ln(52)$, $\text{BMRatio}_i = 0.45$, and $\text{Dow}_i = 1$, the predicted return would be $-0.6483 + 0.1777 \times \ln(52) + 0.0433 \times 0.45 - 0.1641 \times 1 \approx -0.0908 = -9.08\%$.

- ☐ 3.59%.

Incorrect Answer Feedback: Incorrect because this is the result when the dollar amount of the market capitalization is plugged into the estimated model, instead of using the amount expressed in billions of dollars. Further, the answer is incorrectly adjusted to a percentage. The estimated model is $R_i = -0.6483 + 0.1777 \times \text{Size}_i + 0.0433 \times \text{BMRatio}_i - 0.1641 \times \text{Dow}_i$. When $\text{Size}_i = \ln(52,000,000,000)$, $\text{BMRatio}_i = 0.45$, and $\text{Dow}_i = 1$, the expected return would be $-0.6483 + 0.1777 \times \ln(52000000000) + 0.0433 \times 0.45 - 0.1641 \times 1 \approx 3.59$, adding a % sign to get 3.59%.

- ☐ 7.33%.

Incorrect Answer Feedback: Incorrect because this is the expected return for a stock that is not in the DJIA, with a market capitalization of \$52 billion and a book-to-market ratio of 0.45. The estimated model is $R_i = -0.6483 + 0.1777 \times \text{Size}_i + 0.0433 \times \text{BMRatio}_i - 0.1641 \times \text{Dow}_i$. When $\text{Size}_i = \ln(52)$, $\text{BMRatio}_i = 0.45$, and $\text{Dow}_i = 0$, the predicted return would be $-0.6483 + 0.1777 \times \ln(52) + 0.0433 \times 0.45 - 0.1641 \times 0 \approx 0.0733 = 7.33\%$.

17 Multiple Choice 1 / 1 point

The intern's response to Rivera's question regarding the adjusted R^2 measure is:

☐ correct.

Incorrect Answer Feedback: Incorrect because the intern is wrong by stating that adjusted R^2 is nondecreasing in the number of independent variables. When a new independent variable is added, adjusted R^2 can decrease if adding that variable results in only a small increase in R^2 . In fact, adjusted R^2 can be negative, although R^2 is always nonnegative.

☒ incorrect with regard to the nondecreasing feature.

Correct Answer Feedback: Correct because the intern is wrong by stating that adjusted R^2 is nondecreasing in the number of independent variables. When a new independent variable is added, adjusted R^2 can decrease if adding that variable results in only a small increase in R^2 . In fact, adjusted R^2 can be negative, although R^2 is always nonnegative. Adjusted R^2 can be negative as well as decline.

☐ incorrect with regard to its appropriateness for Rivera's model.

Incorrect Answer Feedback: Incorrect because the intern is right that adjusted R^2 would be more appropriate for Rivera's model as there are more than one independent variable. She is wrong by stating that adjusted R^2 is nondecreasing in the number of independent variables. When a new independent variable is added, adjusted R^2 can decrease if adding that variable results in only a small increase in R^2 . In fact, adjusted R^2 can be negative, although R^2 is always nonnegative.

18 Multiple Choice 1 / 1 point

A regression model with the equation $\hat{Y} = 3.14 + 1.59X_1 + 2.65X_2 + 3.58X_3$ is used for prediction. If the forecast values of X_1 , X_2 , and X_3 are 1, -2, and 3, respectively, the predicted value of Y is *closest* to:

☐ 7.03.

Incorrect Answer Feedback: Incorrect because this answer omits the intercept, i.e. $1.59(1) + 2.65(-2) + 3.58(3) = 7.03$.

☒ 10.17.

Correct Answer Feedback: Correct because the predicted value of Y is $3.14 + 1.59(1) + 2.65(-2) + 3.58(3) = 10.17$.

☐ 20.77.

Incorrect Answer Feedback: Incorrect because this answer considers only the magnitude but not the sign of the forecast values of the independent variables, i.e. $3.14 + 1.59(1) + 2.65(2) + 3.58(3) = 20.77$.

19 Multiple Choice 1 / 1 point

Which of the following would *most likely* increase as more independent variables are added to a multiple regression model?

☒ R -squared

Correct Answer Feedback: Correct because, as independent variables are added to the model, R -squared (R^2) will increase or will stay the same, but it will not decrease.

☐ Adjusted R -squared

Incorrect Answer Feedback: Incorrect because adjusted R -squared does not automatically increase when another independent variable is added to a regression. Adjusted R -squared can decrease, stay the same, or increase, while R -squared can only increase or stay the same. Further, in all cases where adjusted R -squared increases, R -squared increases too. However, there are cases where R -squared increases and adjusted R -squared decreases.

☐ The model's goodness of fit

Incorrect Answer Feedback: Incorrect because adding more independent variables may lead to overfitting, reducing the goodness of fit for the model. Overfitting of a regression model is a situation in which the model is too complex, meaning there may be too many independent variables relative to the number of observations in the sample.

20 Multiple Choice 1 / 1 point

An analyst conducts a general linear F -test to investigate the model fit of a multiple regression model. The null hypothesis is $H_0: b_1 = b_2 = 0$, where b_1 and b_2 are the partial regression coefficients of the two independent variables. If the F -statistic is greater than the critical F -value, there is:

- ☐ insufficient evidence to reject the null hypothesis.

Incorrect Answer Feedback: Incorrect because we reject the null hypothesis as the calculated F -statistic exceeds the critical F -value. There is sufficient evidence to indicate that at least one slope (or partial regression) coefficient is different from zero. Since the F -statistic is greater than the critical F -value, there is sufficient evidence to reject the null hypothesis.

- ☒ sufficient evidence to reject the null hypothesis and conclude that at least one partial regression coefficient is different from zero.

Correct Answer Feedback: Correct because we reject the null hypothesis as the calculated F -statistic exceeds the critical F -value. There is sufficient evidence to indicate that at least one slope (or partial regression) coefficient is different from zero.

- ☐ sufficient evidence to reject the null hypothesis and conclude that both partial regression coefficients are different from zero.

Incorrect Answer Feedback: Incorrect because we reject the null hypothesis as the calculated F -statistic exceeds the critical F -value. There is sufficient evidence to indicate that at least one slope (or partial regression) coefficient is different from zero. However, it cannot be concluded that both partial regression coefficients are different from zero.